Numerical Computation of the Fresnel Cosine Integral

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1 Introduction

The goal of this assignment is to numerically compute the Fresnel cosine integral

$$C(x) = \int_0^x \cos\left(\frac{\pi t^2}{2}\right) dt$$

We approximate $C(x)$ using two strategies depending on the magnitude of the input x: a Taylor expansion approximation for small x, and an auxiliary function representation evaluated with Gauss quadrature for large x.

2 Mathematical Background

2.1 Symmetry

We notice:

$$C(-x) = -C(x)$$

This lets us restrict the numerical method to $x \geq 0$.

2.2 Taylor Series (small x)

Expanding $\cos\left(\frac{\pi t^2}{2}\right)$ in its Taylor series and integrating term by term gives

$$C(x) = \sum_{k=0}^{\infty} \frac{(-1)^k \left(\frac{\pi}{2}\right)^{2k} x^{4k+1}}{(2k)!(4k+1)}$$

This series converges for all x, but the number of terms needed for a fixed accuracy grows with x, so it is only efficient for small x.

We do recursive term evaluation, meaning rather than recomputing factorials and powers from scratch at every term, we compute the ratio between consecutive terms s_k and s_{k-1}:

$$\frac{s_k}{s_{k-1}} = -\left(\frac{\pi}{2}\right)^2 x^4 \frac{4k-3}{2k(2k-1)(4k+1)}$$

This allows each new term to be obtained from the previous one, avoiding repeated evaluation of large factorials and powers.

2.3 Auxiliary Functions (large x)

For larger x, the Taylor series requires too many terms, so we use the auxiliary functions representation:

$$C(x) = \frac{1}{2} + f(x)\sin\left(\frac{\pi x^2}{2}\right) - g(x)\cos\left(\frac{\pi x^2}{2}\right)$$

where f and g are given by:

$$f(z) = \frac{1}{\pi\sqrt{2}} \int_0^\infty \frac{e^{-\pi z^2 t^2}}{\sqrt{t}(t^2+1)} dt = \frac{\sqrt{2}}{\pi z} \int_0^\infty \frac{e^{-\pi w^2/2}}{1+(w/z)^4} dw$$

$$g(z) = \frac{1}{\pi\sqrt{2}} \int_0^\infty \frac{\sqrt{t} e^{-\pi z^2 t^2}}{t^2+1} dt = \frac{\sqrt{2}}{\pi z^3} \int_0^\infty \frac{w^2 e^{-\pi w^2/2}}{1+(w/z)^4} dw$$

The change of variables was done so now there is no problem at 0 with the argument function.

The Gaussian factor $e^{-\pi w^2/2}$ decays very fast, so the infinite integration domain can be truncated to a finite interval $[0, W]$ with negligible error, and the resulting integral is evaluated with Gauss-Legendre quadrature.

2.4 Choosing the Cutoff

The two methods are combined by choosing a cutoff value X_{cutoff}: for $0 \le x < X_{\text{cutoff}}$ the Taylor series is used, and for $x \ge X_{\text{cutoff}}$ the auxiliary-function method is used. In my implementation $X_{\text{cutoff}} = 1$.

3 Testing

Correctness is verified by comparing results against the reference implementation `fresnelc` from `FresnelIntegrals.jl` with relative tolerance of $5 \cdot 10^{-11}$.

- **Interval** $(0, 1)$: values ranging from 10^{-6} up to 0.999, this is the Taylor expansion case
- 1 **and beyond**: values ranging from 1 up to 10^5, this is the Gauss quadrature case
- **Negative inputs**: to make sure the negative inputs are correctly computed
- **Exact anti-symmetry**: checks that $-C(-x) = C(x)$ holds
- **Continuity across X_{cutoff}**: checks that the function is continuous at the cutoff point

4 Results

Figure 1 shows $C(x)$ over $x \in [-10, 10]$, generated by `demo.jl`. As expected, $C(x)$ oscillates while damping towards the asymptotes ± 0.5 as $|x| \to \infty$, matching the constant term in the auxiliary-f nction formula.

Figure 2 zooms in on the cutoff point $X_{\text{cutoff}} = 1$, where the implementation switches from Taylor approximation to the quadrature approximation. As we can see, it is continuous.

Overall, the implementation reproduces the reference `FresnelIntegrals.jl` values to within 5×10^{-11} relative error for both positive and negative inputs, with a smooth transition at the cutoff between the two numerical methods.

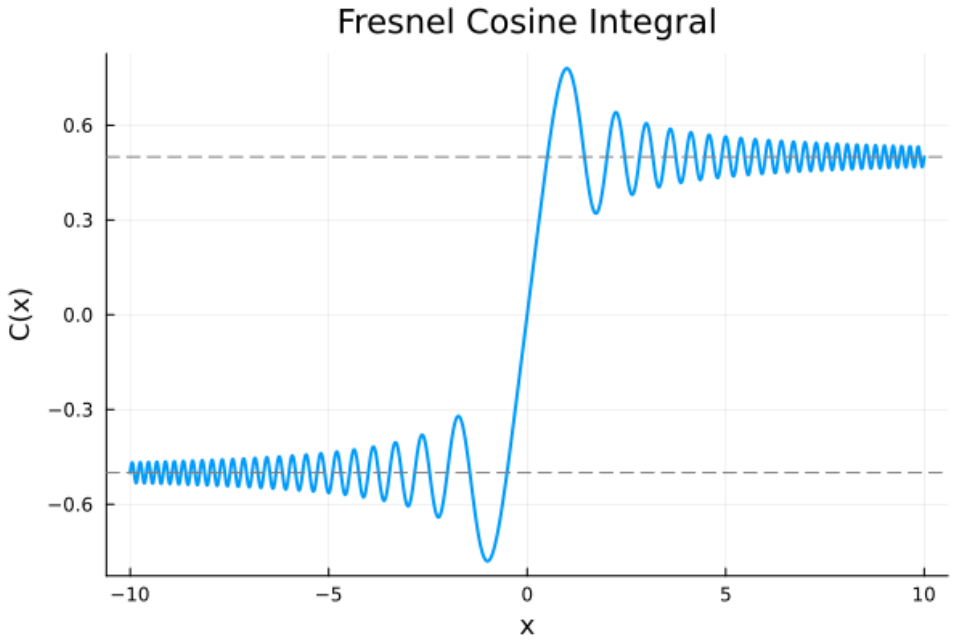


Figure 1: Fresnel cosine integral $C(x)$ over $x \in [-10, 10]$, with asymptotes at ± 0.5 shown as dashed lines

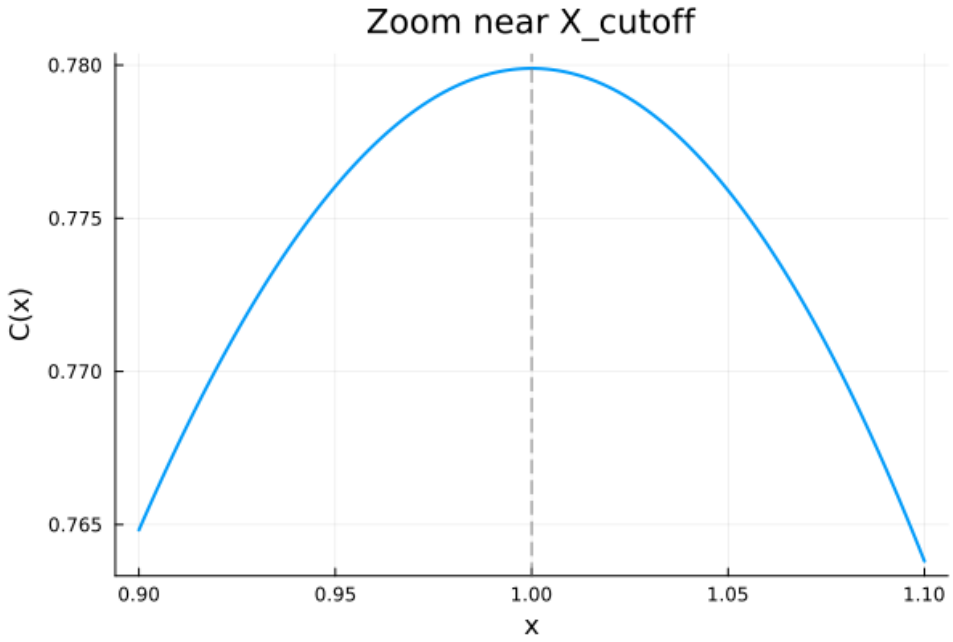


Figure 2: Zoom of $C(x)$ near $x = X_{\text{cutoff}} = 1$ (dashed line), where the method switches from the Taylor approximation to the quadrature approximation with auxiliary functions